EXTENSION WORK

Unlike diamond and graphite C_{60} fullerene does dissolve in some solvents; only relatively weak intermolecular forces of attraction have to be broken for it to dissolve.

C_{60} fullerene is not as hard as diamond. It does not take as much energy to break the intermolecular forces of attraction in C_{60} fullerene compared to breaking the strong covalent bonds in diamond.

C_{60} fullerene does not conduct electricity. Although all the carbon atoms in C_{60} only form three bonds, the fourth electron on each atom can only move around *within* each C_{60} molecule; the electrons cannot jump from molecule to molecule.

CHAPTER QUESTIONS

You will need to use the Periodic Table in Appendix A on page 320.

SKILLS CRITICAL THINKING 

1 State whether each of the following compounds is ionic or covalent:

a MgO b CH_3Br c H_2O_2

d $FeCl_2$ e NaF f HCN

SKILLS INTERPRETATION 

2 a What is meant by a covalent bond? How does this bond hold two atoms together?

b Draw dot-and-cross diagrams (showing outer shell electrons only) to show the covalent bonding in

i methane (CH_4)

ii hydrogen sulfide (H_2S)

iii phosphine (PH_3)

iv silicon tetrachloride ($SiCl_4$)

c Draw dot-and-cross diagrams (showing outer shell electrons only) to show the covalent bonding in

i O_2

ii N_2

3 Draw dot-and-cross diagrams (showing outer shell electrons only) to show the covalent bonding in

a ethane (C_2H_6) b ethene (C_2H_4) c chloroethane (CH_3CH_2Cl)

SKILLS REASONING

4 Explain why carbon dioxide sublimes at –78.5 °C but diamond sublimes at around 4000 °C.

5 Hexane has the formula C_6H_{14}. It is a liquid at room temperature.

a Explain whether hexane has a simple molecular or giant structure.

b Explain whether you would expect pentane (C_5H_{12}) to have a higher or lower boiling point than hexane.

c Explain whether or not you would expect hexane to conduct electricity.



6 Explain the following in terms of structure and bonding:

a diamond is harder than graphite

b C_{60} fullerene has a lower melting point than graphite

c graphite conducts electricity

d diamond does not conduct electricity.

SKILLS ANALYSIS

8

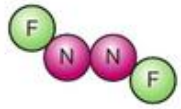
7 The table below gives details of the boiling temperatures of some substances made of covalent molecules. Arrange these substances in increasing order of the strength of their intermolecular forces and explain your order.

Substance	Boiling point/°C
ammonia	–33
ethanamide	221
ethanol	78.5
hydrogen	–253
phosphorus trifluoride	–101
water	100

(Don't panic if you don't recognise some of the names. The substances could also have been labelled A, B, C, D, E and F.)

SKILLS REASONING

8 The compound N_2F_2 has the structure:



a Explain whether you would expect the bond between the nitrogen atoms to be a single, double or triple bond.

SKILLS INTERPRETATION

b Draw a dot-and-cross diagram for N_2F_2 showing outer electrons only.

SKILLS REASONING

9 a Draw a dot-and-cross diagram (showing outer electrons only) to show the covalent bonding in BCl_3.

9

b BCl_3 is sometimes described as an *electron-deficient* compound. Explain what you think that means.